

ENGG2112 Multi-Disciplinary Engineering

Semester 2, 2026
Week 1

Professor T. J. Lim
Professor Mary Lou Maher



THE UNIVERSITY OF
SYDNEY

Celebrating **175** *years*

CRICOS 00026A TEQSA PRV12057



Unit Overview

*Teaching staff, learning activities, assessments,
learning objectives, etc.*

Unit Coordinators/Lecturers

Prof. T. J. Lim

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J12 Level 5



Prof. Mary Lou Maher

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J12 Level 4



Tutors

- Dr Ahsan Rizvi
- Dr Elyas Shamsabadi
- Ms Clarissa Miranda Pena
- Mr Chenyu Xu
- Mr Colin Zuhlke
- Mr Jasper Wang
- Mr Jianheng Lan
- Mr Pratul Singh Raghava
- Mr Yunfeng Zhu

Lectures and Tutorials

- **Lectures** on Mondays, 4.00 to 6.00pm, F11 Chemistry Bldg, LT1
- **Tutorials** in 11 groups:
 - Group 1: Wed 9.00 to 11.00am, Merewether 158
 - Group 2: Thu 3.00 to 5.00pm, Carlaw 351
 - Group 3: Thu 5.00 to 7.00pm, Civil 304
 - Group 4: Fri 5.00 to 7.00pm, Law Annex 442
 - Group 5: Fri 5.00 to 7.00pm, Law Annex 346
 - Group 6: Tue 3.00 to 5.00pm, Mechanical S220
 - Group 7: Tue 8.30 to 10.30am, BHB 3190
 - Group 8: Wed 9.00 to 11.00am, BHB 2280
 - Group 9: Tue 11.00am to 1.00pm, Civil 304
 - Group 10: Fri 9.00 to 11.00am, Mechanical S220
 - Group 11: Thu 1.00 to 3.00pm, BHB 2240



Tutorials

- Starts **this week!**
- Find one other person you want to be grouped with in the first two tutorials.
- Submit names to your tutor by end of second tutorial (10 to 14 August).
- Two pairs will be put together in a project team with more than one engineering stream.
- You will be buddies for the semester, consulting each other not just in your project but in all other class activities.



Shift in Focus

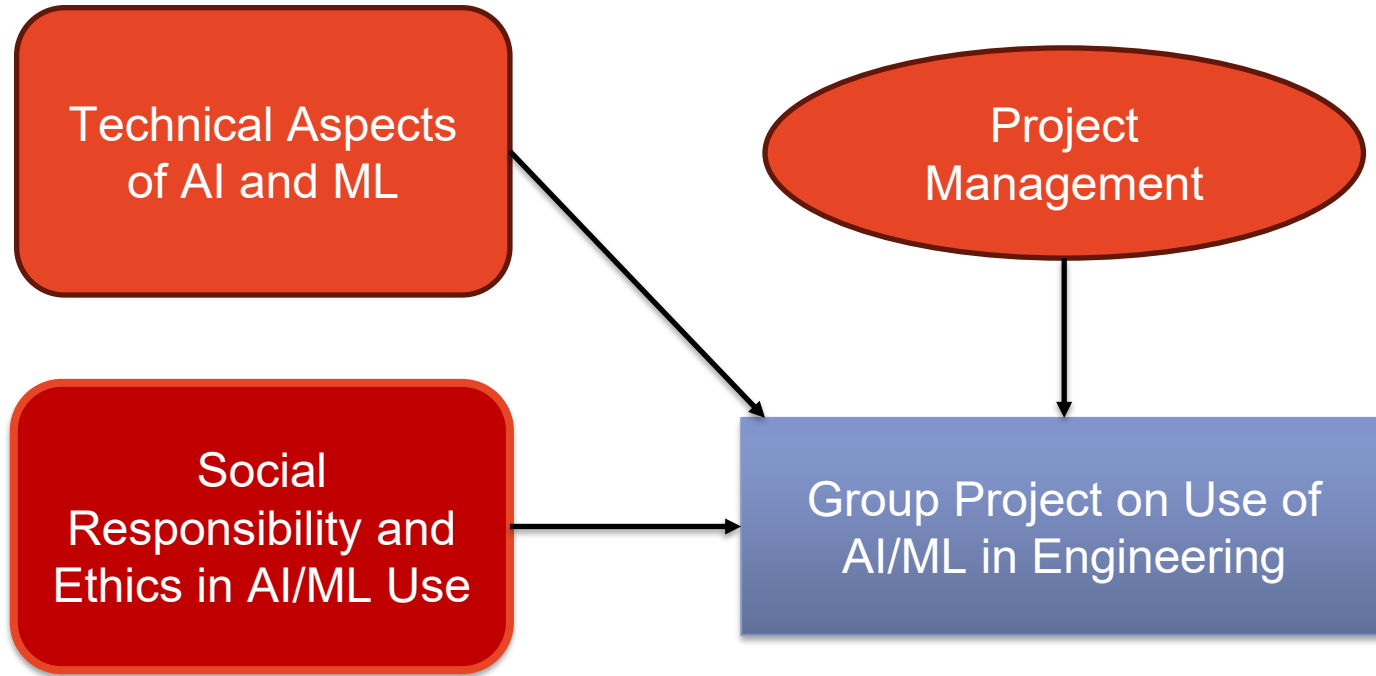
- **Previous** versions of ENGG2112 – learn technical aspects of AI/ML, then use AI/ML in a group project.
- **New** version of ENGG2112 from this semester – learn **socio**-technical aspects of AI/ML, especially **generative AI**, then **use AI for personalized learning of new material** as well as in a group project.

WHY?

We are living through an industrial revolution

- The way we live, work and play is changing rapidly
- We need to know how to leverage our human abilities in collaboration with AI to improve society
- As engineers, you need to develop an intuitive sense of what AI is and is not, based on what we will cover in this semester

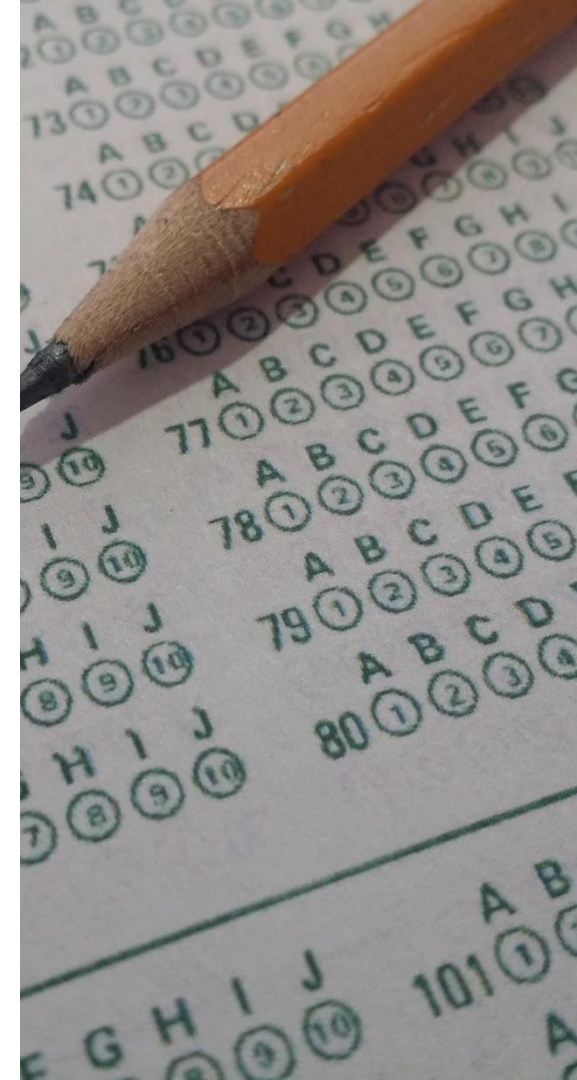
What is in ENGG2112



Assessments

- **Individual Tasks** – Use AI as a learning and project partner
 - Week 3: AI conversation transcript #1 (open) – 5%
 - Week 5: AI conversation transcript #2 (open) – 5%
 - Week 5: Explain learnt topic to other project members (secure) – 5%
- **Group Project** – Use AI as part of the solution to an engineering problem
 - Week 6: Proposal (open) – 10%
 - Week 12: Final Report (open) – 15%
 - Week 13: Q&A/Oral Exam (secure) – 20%
- **Final Exam (secure)** – 40%
- **Tutorial Attendance (hurdle)** – at least 80% required

Total secure = 65%, Total open = 35%



Attendance

- Compulsory attendance in at least 80% of tutorials
 - No need for Spec Con if you're going to miss up to two tutorials.
 - If you miss three or more tutorials without SC, it will be an Absent Fail (AF) regardless of the marks accumulated over the semester.
- Lecture attendance is not compulsory but strongly encouraged
 - If you don't attend the lecture in-person, at least watch the recording.

Learning Objectives

- Understand basics of machine learning and AI, and how they can be used to solve complex problems.
- Develop ability to use generative AI to accelerate learning and increase productivity.
- Understand safety, ethics and regulatory aspects of AI.
- Define an engineering problem and iterate through multiple solutions to the problem.
- Manage a team project successfully.
- Present updates and results, write reports.

AI and Machine Learning

Tools for Solving Problems

A Toolbox for Solving Complex Problems

- **New this semester:** Teaching you how to use generative AI in productive and safe ways.
 - Both AI for learning and AI for engineering
 - How should humans interact with AI in the age of AI?
- We're using this unit as an opportunity to introduce you, engineers of the future, to this toolbox of the future.
- You will learn through using AI to learn new topics – both seen and unseen in lectures – and then using AI in your project.
- The **team project** requires you to exercise your uniquely human creativity and sense of purpose, connecting technical solutions to real problems.

Issues and Open Questions

- AI is bound to replace routine human mental labour but will it replace human ingenuity?
 - Where is **the labour market** headed in the age of AI?
- Is AI growth going to deplete natural resources (energy and water) and will the gains arising from AI be worth the **environmental costs** of mass AI?
 - Are the AI pessimists wrongly assuming that energy efficiency will not improve in the future?
- As AI gets smarter and is used more routinely, are humans destined to become “dumber”?
 - **How humans interact with AI** matters a lot. We humans have the agency to ensure that AI interactions benefit us as humans!
- If AI is used in critical, even **life-and-death**, scenarios, who is responsible if something goes wrong?
- How does society **protect vulnerable people** from AI outputs generated for nefarious purposes?
- How do we draw the line between **productive use of AI** and unethical or illegal use of AI?
 - Is it ever right to claim AI outputs as our own work? What if AI produces a first cut and we carefully review and revise it?

Good Practices When Using AI

- Be ready to think and challenge. To do this, you need to **have an opinion and a goal**.
- Plan on having **a conversation**, not a one-off search query.
- **Verify information** provided by AI – first ask for sources, then read them.
 - AI hallucination continues to be a thing.
- Give prompts that contain detail, e.g. **role playing** (“You are a patient advisor and I am seeking your help on study habits”), **your own understanding** of the problem (“I think that the variable t in signals and systems always refers to continuous time. Am I right?”), **precise descriptions** (“I need a Python function that replaces all instances of positive values in the attached Excel with +”).
 - The more context you can give AI, the better its response
- In a long conversation, remember that AI’s context window is limited. Ask it to summarize the earlier conversation so that the context is refreshed.



Open Claude Desktop

Formative Assessments

- We will be introducing formative low-weight assessments to encourage you to engage with AI productively.
- Assignment 1 asks that you use AI to learn more about the material introduced in our first three lectures.
- Assignment 2 asks that you use AI to learn about a topic that you don't know much about and then explain the topic to your project team.
- 15% of total marks

Technical Underpinnings of AI

- We will discuss these from Week 3
- Long and interesting history with technical roots in topics like statistical signal processing, Bayesian inference, stochastic detection and estimation theory, adaptive systems, etc.
- Powerful parallel computing hardware has enabled the realisation of computational learning methods based on theory or heuristics.
 - Question: how do we make use of them intelligently?
 - This is a question we would like you to consider deeply...

What is AI/ML?



Definition of artificial intelligence (?)

Intelligence = *ability to take actions that lead to desirable outcomes with an adequately high probability in an environment that frequently results in a state of nature not previously encountered*

This means the following systems are examples of **artificial** intelligence:

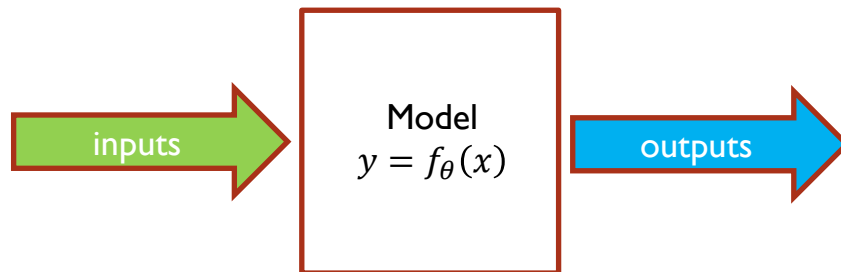
- **Automatic cruise control in a car** – able to regulate speed on any road under any conditions
- **Noise cancelling headphones** – works to some acceptable degree with any noise and any audio signal
- **Power system** – provides required power to an uncertain number of consumers at any time
- **Google Maps** – able to find a route from any place to any place at any time and estimate travel time
- **Social media recommendations** – able to suggest people/places/etc. to check out based on individual data
- **ChatGPT** – able to respond with near-human (or super-human) intelligence to any question on any topic

What Does AI Mean Today?

- At its most basic, an AI is nothing more than the implementation of a **mapping** from a set of inputs to one or more outputs
 - Human intelligence is also the same – in response to our sensory inputs, we produce thoughts, words, muscle activations, etc.
- AI, therefore, encompasses many of the “old” or “traditional” engineered systems, which can be construed to be making an automated decision, by whatever means.
 - But this definition is a bit too broad and AI currently refers only to **computational** methods for arriving at these decisions.
 - Even this narrower definition is quite broad, since it includes all programmed systems such as if/then/elseif **rules-based systems**.
- A lot of what we mean by AI these days is really Machine Learning (ML)
 - ML refers to a computational machine that ingests data and **learns a model** that links input data (features) to output data (labels). This is what we focus on in ENGG2112.

Abstract view of Machine Learning: Model

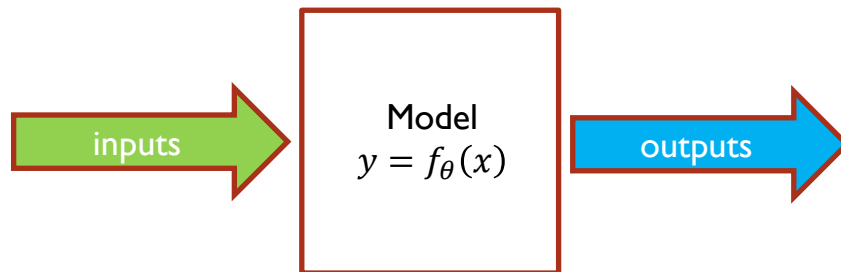
- First, define a model linking inputs to output(s)
- This model has a set of parameters denoted θ
- The critical task of ML is to find the **best values** for the parameters θ using available data
- This is accomplished in the training phase



Abstract view of Machine Learning: Model

Examples:

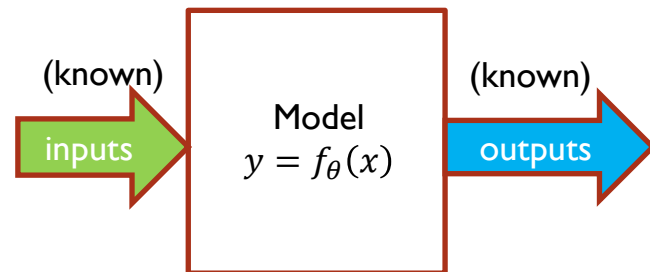
- **Input:** Optical signals into C3PO's eyes. **Output:** Recognized object (R2D2, Luke, etc.)
- **Input:** Sensors around car. **Output:** Steering, brake, accelerator signals
- **Input:** Audio signals into Google Home. **Output:** Recognized command
- **Input:** Sensors on robot. **Output:** Signals to each leg to control movement.



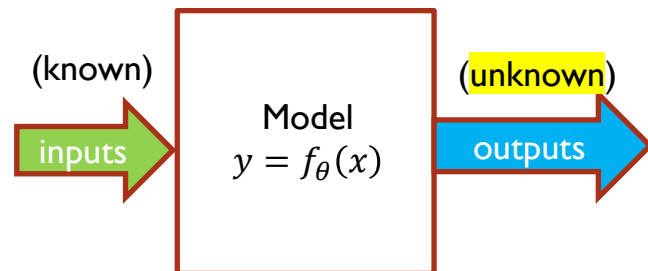
Abstract view of machine learning: Training/learning

- In the **training phase**, we use known inputs and outputs (supervised learning) to find θ .
- In unsupervised learning, the outputs are not known and therefore θ can't be found. Instead, **objective is to discover patterns in the data.**
- There is a third class of learning called reinforcement learning which relies on feedback from the environment in response to an action taken by the machine to discover the model parameters. Eg a scoring function

Supervised Learning

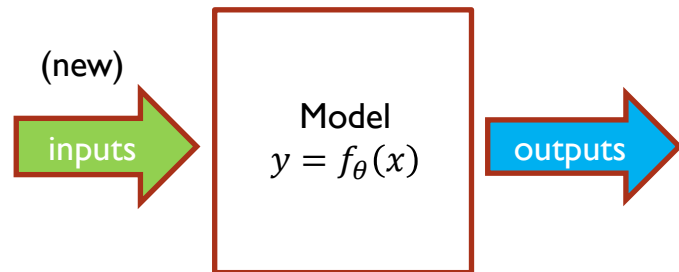


Unsupervised Learning



Abstract view of machine learning: Deployment

- After training, we have the “best” setting for θ and are now ready to deploy the model. This stage is also called “generalisation”.
- For instance:
 - A human driver would train a self-driving car. After training, it would (hopefully) be ready to drive itself.
 - C3PO would be shown many pictures of R2D2 and Luke for training. After training, he would be able to tell them apart.
 - Google Home would be trained to recognize human speech through many example phrases spoken by different people. After training, it's able to recognize speech spoken by anyone.



Keys to the Rapid Development of ML

1. Acceptance That Explanation is Not Always Necessary

Engineers and scientists traditionally sought to understand model functionality.

ML shows that sometimes, we only need to know it works reliably, even without full explanations.

Example: In smartphones, users care only that facial recognition works accurately, not how it does.

2. Advances in Algorithms and Computational Power

In the 1990s, neural networks and deep learning were possible but computationally impractical.

Today, GPUs and TPUs enable massive parallel computing, making complex models trainable in hours. For example, image recognition that once needed supercomputers can now be done on a standard laptop.

3. Explosion of Data Availability

ML models need large datasets to train effectively.

In the 90s, data collection was limited, but today, IoT, smartphones, and online platforms generate vast amounts of data. Example: Self-driving cars—modern ML models continuously improve as they process millions of miles of real-world driving data.

Traditional Methods Still Matter - Shouldn't Discard Them

1. If a Traditional Model Works, Keep It

Many engineering problems have established analytical models that are efficient and reliable. ML should not replace these methods unless it offers a clear advantage. For example, in structural engineering, finite element analysis (FEA) is understood and reliable, so ML may not provide significant value.

2. Engineers Must Still Learn Traditional Methods

Many ML models are black boxes, meaning they lack the interpretability of traditional models. Understanding fundamental engineering principles is crucial for verifying ML outputs. Example: ML can predict turbulence patterns in fluid mechanics, but engineers still need to understand Navier-Stokes equations to validate results.

3. Hybrid Approaches Are the Future

The best engineering solutions often combine traditional models with ML. Example: Predictive maintenance in aviation—traditional physics-based failure models combined with ML anomaly detection create more accurate and reliable predictions.

Project Work

Aims of the Group Project

- Learn to work in a team of diverse people
 - There's a separate set of slides on project management that you can use as guidance on how to run your project
- Learn to identify goals in a project that are ambitious but achievable
- Learn the importance of a workplan and regular project updates and meetings
- Leverage the project to promote your expertise and achievements on LinkedIn
- Learn how to use AI as an additional team member (or two)
- Learn how AI can be used to solve interesting engineering problems

Career Preparation and Employability

- Your ultimate aim is to land a fulfilling job. How do you maximize your chances of doing that?
- Throughout the project, work on discovering your strengths and weaknesses – how do you amplify those strengths and overcome those weaknesses?
 - Teamwork and collegiality
 - Written and verbal communications
 - Empathy and relating to others
 - Work attitude
 - Attention to detail
 - Technical competency
- Even if you're “only” in second year, it's not too soon to think about life after graduation. Start preparing for it in this unit of study.

Broad Project Objectives

- “AI for Good” – we will provide a few tutorial-style papers to get you started.
- Technical components of project – use AI/ML to solve a problem chosen by your team.
- Non-technical components – understand how your proposed solutions can impact society both positively and negatively.
- AI as a learning accelerator – learn to use AI to your advantage.
- Multi-disciplinary aspects – appreciate that most engineering projects are not confined to one well-defined domain; and that AI/ML is applicable to/disrupting all fields of engineering.

Examples of Projects

Drones for Surveillance and Tracking

- Police and military may need to track vehicles or people on the ground using drones
- *Technical problem*: How can multiple drones be used to improve performance over using just one drone? What are the cost-benefit trade-offs?
- *Other questions*: What are the consequences of identification errors? Are these ethically acceptable? Are there legal implications?



Cybersecurity

- Malware and viruses spread from one computer to another and ideally, should be stopped early.
- *Technical question*: Can we use AI operating on network parameters to detect the spread of malware and viruses?
- *Other questions*: Who operates and owns the AI? How would the spread be stopped? What are the costs of stopping the spread?



Detecting Driver Fatigue

- Driver fatigue is a major cause of road accidents. Early detection can save lives.
- *Technical question*: Using onboard cameras and other sensors, can AI be used to detect the early onset of driver fatigue?
- *Other questions*: Will this increase cost substantially? Can we quantify the benefits of this technology in terms of lives saved? What level of false alarm and missed detections can we accept?



Negative Examples...

- Prediction of stock market index in the future based on past without discussion of how it will be used and the cost of inaccurate predictions, etc.
- Classification of water as potable or not, based on chemical readings – do we need ML or AI for this?
- Prediction of bushfire at a certain location based on current weather conditions and knowledge of where bushfires are burning – why would AI/ML work better than conventional methods?
- Diagnosis of diabetes based on non-invasive measurement of vital signs – accuracy is low relative to simple blood test.

Q&A – what do you want to know?



<https://app.sli.do/event/tQrbCkdiYkbtHg2hXoN2ji>

What's Coming Up?

Next Week's Class

Week 2 Preview

Short Survey

